Module 41.10 (Hinglish): Comparative Study: Sequence Processing with RNNs vs Spatial Convolutions with CNNs

1. CNN aur RNN ka Comparison

Deep Learning me do major neural network families hoti hain:

1. **Convolutional Neural Networks (CNN):** Spatial data (Images, Grid data) ke liye best hote hain.
2. **Recurrent Neural Networks (RNN):** Temporal data (Text, Audio, Sequences) ke liye best hote hain.

Lekin kya ek RNN image classification kar sakta hai? Haan! **MNIST Handwritten Digit Dataset (28×28)** ko hum dono models se process karke compare kar sakte hain (jaise Day 41 reference image `9 Handwritten Digit Classification using CNN & RNN.png` me show kiya gaya hai).

---

2. Notebook-Friendly Hand-Drawn Architecture Diagram

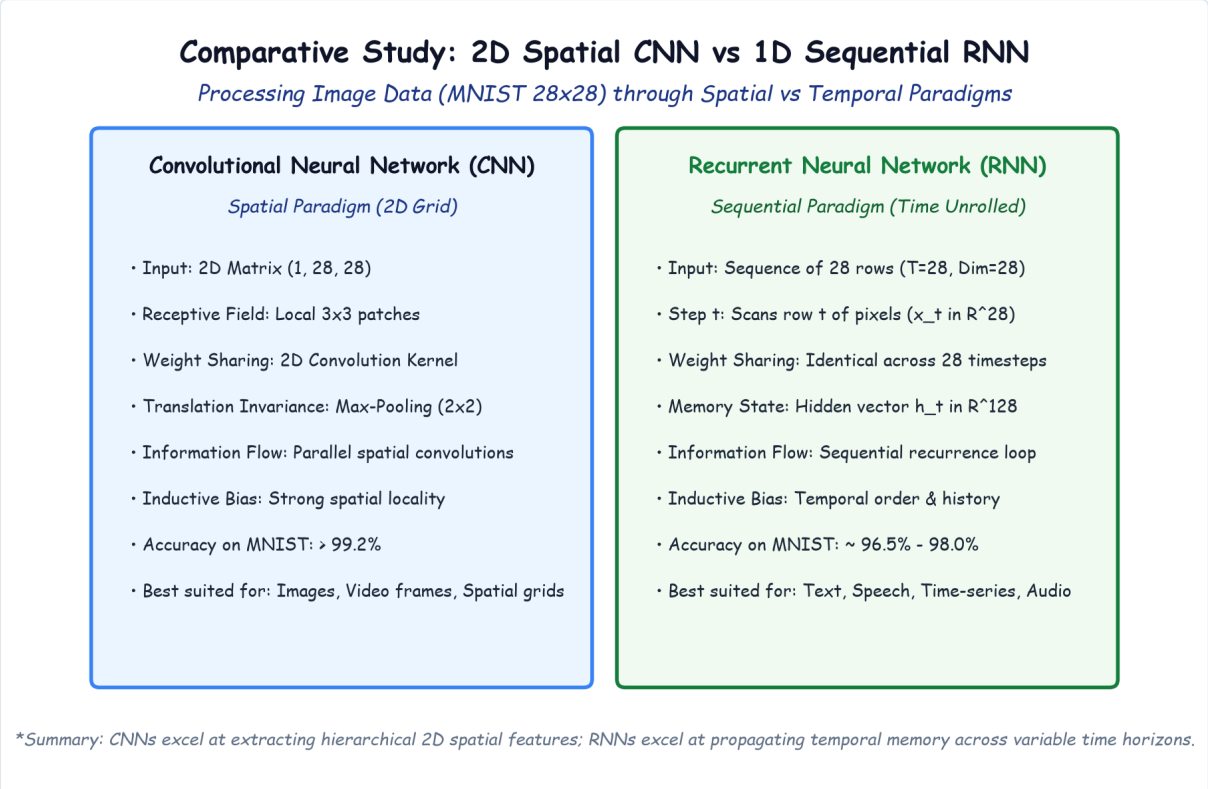


Figure 41.10: MNIST dataset par 2D Spatial Convolution vs 1D Sequential Recurrence ka clear visual comparison: CNN 2D grid patches scan karta hai jabki RNN 28 rows ko 28 time steps ki tarah scan karta hai.

---

3. Image ko Process Karne ke Do Alag Tareeqe

Tareeqa 1: CNN (2D Spatial Convolutions)

- Image ko ek 2D surface maanta hai ($(1, 28, 28)$).
- Chhote filters (3×3) horizontal aur vertical dono directions me ek sath move karte hain.
- **Translation Invariance:** Digit thoda left ya right shift ho jaye, CNN fir bhi use easily pehchan leta hai.
- **Accuracy:** $> 99.2\%$.

Tareeqa 2: RNN (1D Row-by-Row Sequence)

- Image ko 28 time-steps ka ek sequence bana diya jata hai ($T = 28$).
- Step $t = 1$ par Row 1 (28 pixels) input hoti hai.
- Step $t = 2$ par Row 2 input hoti hai aur pichli row ki memory $\mathbf{h}_1$ use hoti hai.
- Aakhiri step $t = 28$ par final hidden state $\mathbf{h}_{28}$ se digit class (0 se 9) predict hoti hai.
- **Accuracy:** $\approx 96.5\% - 98.0\%$.

---

4. Key Differences Table

Feature	CNN	RNN
Input Type	2D Spatial Grid (C, H, W)	1D Time Sequence (T, dim)
GPU Parallelism	**Bohot High** (Saare patches ek sath compute hote hain)	**Low** (Step t ko step $t-1$ ka wait karna padta hai)
Best suited for	Computer Vision (Images, Video frames)	Natural Language (Text, Speech, Audio)
MNIST Performance	$> 99.2\%$	$\approx 97\%$